D1

File Edit View Options Transfer Script Tools Window Help

P1 P2 A1 A2 S1 D1 D2 S2 R4 R3 R2 R1 I3 I2 I1 Google-DNS

```
D1(config)#  ip dhcp excluded-add 10.2.1.1 10.2.1.99
D1(config)#ip dhcp excluded-add 10.2.1.200 10.2.1.254
D1(config)#!
D1(config)#ip dhcp pool vlan10
D1(dhcp-config)#  network 10.2.1.0 255.255.255.0
D1(dhcp-config)#  default-router 10.2.1.254
D1(dhcp-config)#config t
                 ^
% Invalid input detected at '^' marker.

D1(dhcp-config)#exit
D1(config)#exit
D1#config t
Enter configuration commands, one per line.  End with CNTL/Z.
D1(config)#end
D1#config t
Enter configuration commands, one per line.  End with CNTL/Z.
D1(config)#int e1/0
D1(config-if)#   switchport mode access
D1(config-if)#   switchport access vlan 100
D1(config-if)#   do show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Et2/0, Et2/1, Et2/2, Et2/3
                                                Et3/0, Et3/1, Et3/2, Et3/3
10   homevlanApr                      active
20   DNS                              active
100  VLAN0100                         active    Et1/0
200  VLAN0200                         active
1002 fddi-default                     act/unsup
1003 token-ring-default               act/unsup
1004 fddinet-default                  act/unsup
1005 trnet-default                    act/unsup
D1(config-if)#
```

Ready   Telnet: 192.168.192.12 35, 15 35 Rows, 118 Cols Xterm CAP NUM

D2

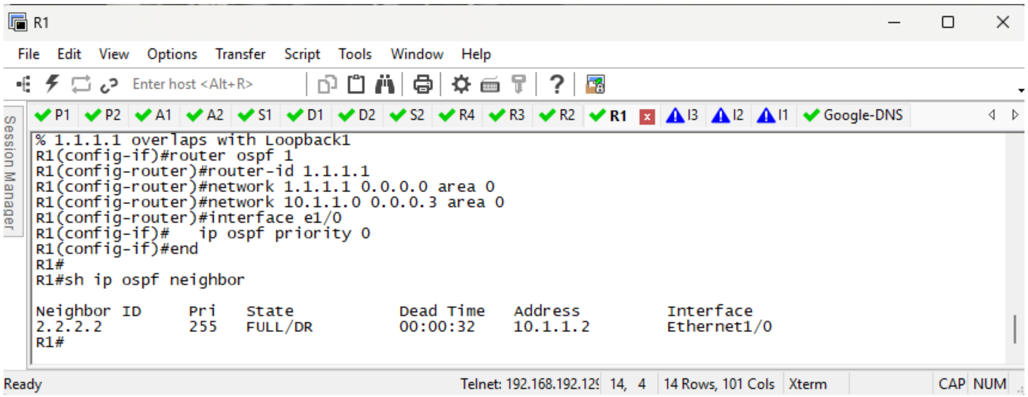
File Edit View Options Transfer Script Tools Window Help

P1 P2 A1 A2 S1 D1 D2 S2 R4 R3 R2 R1 I3 I2 I1 Google-DNS

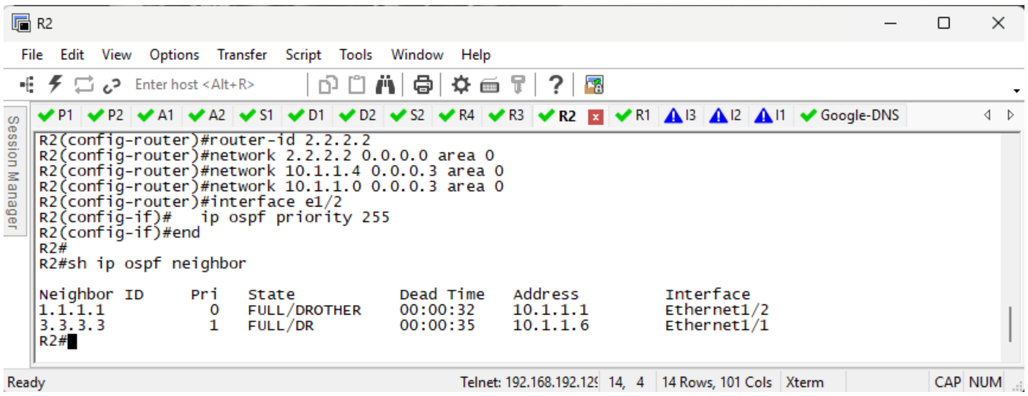
```
D2>en
Password:
D2#
D2#config t
Enter configuration commands, one per line.  End with CNTL/Z.
D2(config)#  ip dhcp excluded-add 10.2.1.1 10.2.1.199
D2(config)#ip dhcp excluded-add 10.2.1.250 10.2.1.254
D2(config)#
D2(config)#ip dhcp pool vlan10
D2(dhcp-config)#  network 10.2.1.0 255.255.255.0
D2(dhcp-config)#  default-router 10.2.1.254
D2(dhcp-config)#  end
D2#
D2#config t
Enter configuration commands, one per line.  End with CNTL/Z.
D2(config)#int e1/0
D2(config-if)#   switchport mode access
D2(config-if)#   switchport access vlan 20
D2(config-if)#   do show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Et2/0, Et2/1, Et2/2, Et2/3
                                                Et3/0, Et3/1, Et3/2, Et3/3
10   homevlanApr                      active
20   DNS                              active    Et1/0
100  VLAN0100                         active
200  VLAN0200                         active
1002 fddi-default                     act/unsup
1003 token-ring-default               act/unsup
1004 fddinet-default                  act/unsup
1005 trnet-default                    act/unsup
D2(config-if)#
```

Ready   Telnet: 192.168.192.12 35, 15 35 Rows, 118 Cols Xterm CAP NUM

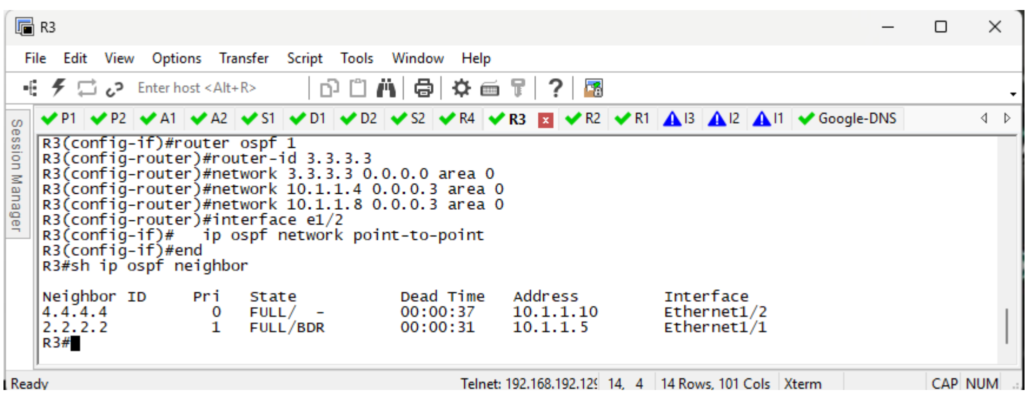
CHECKING OF OSPF NEIGHBOR FOR R1 TO R4



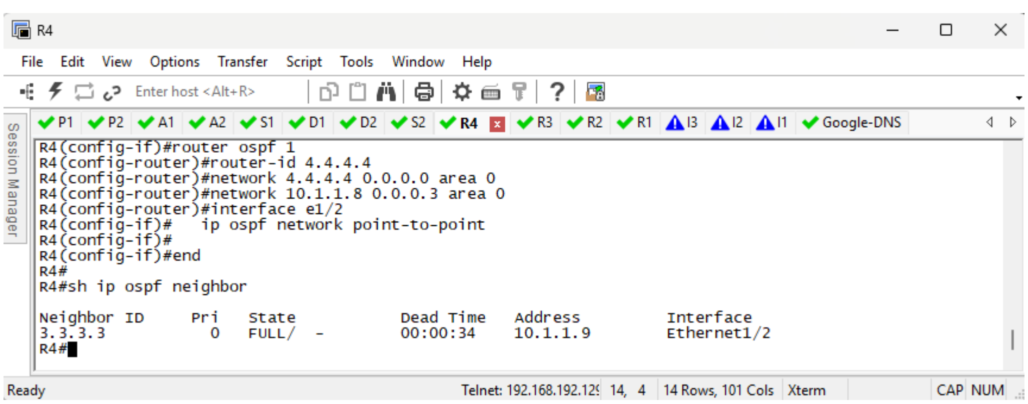
```
% 1.1.1.1 overlaps with Loopback1
R1(config-if)#router ospf 1
R1(config-router)#router-id 1.1.1.1
R1(config-router)#network 1.1.1.1 0.0.0.0 area 0
R1(config-router)#network 10.1.1.0 0.0.0.3 area 0
R1(config-router)#interface e1/0
R1(config-if)#   ip ospf priority 0
R1(config-if)#end
R1#
R1#sh ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
2.2.2.2         255   FULL/DR         00:00:32    10.1.1.2        Ethernet1/0
R1#
```



```
R2(config-router)#router-id 2.2.2.2
R2(config-router)#network 2.2.2.2 0.0.0.0 area 0
R2(config-router)#network 10.1.1.4 0.0.0.3 area 0
R2(config-router)#network 10.1.1.0 0.0.0.3 area 0
R2(config-router)#interface e1/2
R2(config-if)#   ip ospf priority 255
R2(config-if)#end
R2#
R2#sh ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.1           0   FULL/DROTHER    00:00:32    10.1.1.1        Ethernet1/2
3.3.3.3           1   FULL/DR         00:00:35    10.1.1.6        Ethernet1/1
R2#
```



```
R3(config-if)#router ospf 1
R3(config-router)#router-id 3.3.3.3
R3(config-router)#network 3.3.3.3 0.0.0.0 area 0
R3(config-router)#network 10.1.1.4 0.0.0.3 area 0
R3(config-router)#network 10.1.1.8 0.0.0.3 area 0
R3(config-router)#interface e1/2
R3(config-if)#   ip ospf network point-to-point
R3(config-if)#end
R3#sh ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
4.4.4.4           0   FULL/  -        00:00:37    10.1.1.10       Ethernet1/2
2.2.2.2           1   FULL/BDR        00:00:31    10.1.1.5        Ethernet1/1
R3#
```



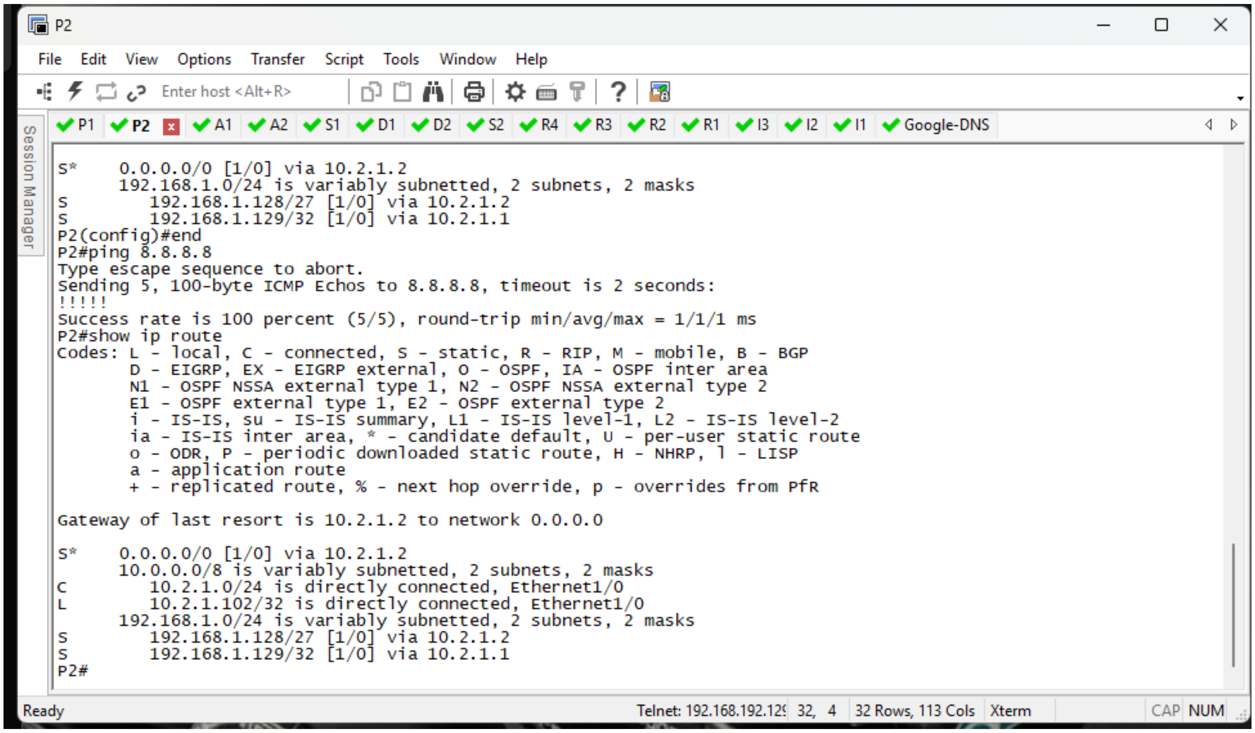
```
R4(config-if)#router ospf 1
R4(config-router)#router-id 4.4.4.4
R4(config-router)#network 4.4.4.4 0.0.0.0 area 0
R4(config-router)#network 10.1.1.8 0.0.0.3 area 0
R4(config-router)#interface e1/2
R4(config-if)#   ip ospf network point-to-point
R4(config-if)#
R4(config-if)#end
R4#
R4#sh ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
3.3.3.3           0   FULL/  -        00:00:34    10.1.1.9        Ethernet1/2
R4#
```

TASK3: EIGRP AND OSPF REDISTRIBUTION:

```
Neighbor ID     Pri   State           Dead Time   Address         Interface
3.3.3.3           0   FULL/  -        00:00:37    10.1.1.9        Ethernet1/2
R4#
R4# config T
Enter configuration commands, one per line.  End with CNTL/Z.
R4(config)#router eigrp 100
R4(config-router)#redistribute ospf 1 metric 10000 100 255 1 1500
R4(config-router)# exit
R4(config)#router ospf 1
R4(config-router)#redistribute eigrp 100 subnets
R4(config-router)#end
R4#sh ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
3.3.3.3           0   FULL/  -        00:00:31    10.1.1.9        Ethernet1/2
R4#show ip EIGRP neighbors
EIGRP-IPv4 Neighbors for AS(100)
H   Address                 Interface              Hold Uptime   SRTT   RTO  Q  Seq
                                                   (sec)         (ms)       Cnt Num
1   10.1.4.6                Et1/0                    13 00:15:35 1281  5000  0  25
0   10.1.4.10               Et1/1                    14 00:15:35    8   100  0  20
R4#
```

TASK 4: (CHECKING: ping 8.8.8.8 and show ip route)

```
P1#
P1#  ping 8.8.8.8
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 8.8.8.8, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
P1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is 10.2.1.1 to network 0.0.0.0

S*    0.0.0.0/0 [1/0] via 10.2.1.1
      10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        10.2.1.0/24 is directly connected, Ethernet0/0
L        10.2.1.101/32 is directly connected, Ethernet0/0
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
S        192.168.1.128/27 [1/0] via 10.2.1.1
S        192.168.1.130/32 [1/0] via 10.2.1.2
P1#  ping 8.8.8.8
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 8.8.8.8, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
P1#
```



```
S*    0.0.0.0/0 [1/0] via 10.2.1.2
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
S        192.168.1.128/27 [1/0] via 10.2.1.2
S        192.168.1.129/32 [1/0] via 10.2.1.1
P2(config)#end
P2#ping 8.8.8.8
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 8.8.8.8, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
P2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is 10.2.1.2 to network 0.0.0.0

S*    0.0.0.0/0 [1/0] via 10.2.1.2
      10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        10.2.1.0/24 is directly connected, Ethernet1/0
L        10.2.1.102/32 is directly connected, Ethernet1/0
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
S        192.168.1.128/27 [1/0] via 10.2.1.2
S        192.168.1.129/32 [1/0] via 10.2.1.1
P2#
```

R1

File Edit View Options Transfer Script Tools Window Help

P1 P2 A1 A2 S1 D1 D2 S2 R4 R3 R2 R1 I3 I2 I1 Google-DNS

```
2.2.2.2           255   FULL/DR         00:00:31    10.1.1.2        Ethernet1/0
R1#ping 8.8.8.8
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 8.8.8.8, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is 208.8.8.4 to network 0.0.0.0

B*    0.0.0.0/0 [20/0] via 208.8.8.4, 00:16:23
      1.0.0.0/32 is subnetted, 1 subnets
C        1.1.1.1 is directly connected, Loopback1
      2.0.0.0/32 is subnetted, 1 subnets
O        2.2.2.2 [110/11] via 10.1.1.2, 00:32:44, Ethernet1/0
      3.0.0.0/32 is subnetted, 1 subnets
O        3.3.3.3 [110/21] via 10.1.1.2, 00:32:44, Ethernet1/0
      4.0.0.0/32 is subnetted, 1 subnets
O        4.4.4.4 [110/31] via 10.1.1.2, 00:32:44, Ethernet1/0
      8.0.0.0/32 is subnetted, 1 subnets
B        8.8.8.8 [20/0] via 208.8.8.4, 00:14:20
      10.0.0.0/8 is variably subnetted, 8 subnets, 3 masks
C        10.1.1.0/30 is directly connected, Ethernet1/0
L        10.1.1.1/32 is directly connected, Ethernet1/0
O        10.1.1.4/30 [110/20] via 10.1.1.2, 00:32:44, Ethernet1/0
O        10.1.1.8/30 [110/30] via 10.1.1.2, 00:32:44, Ethernet1/0
O E2     10.1.4.4/30 [110/20] via 10.1.1.2, 00:23:45, Ethernet1/0
O E2     10.1.4.8/30 [110/20] via 10.1.1.2, 00:23:45, Ethernet1/0
O E2     10.2.1.0/24 [110/20] via 10.1.1.2, 00:23:45, Ethernet1/0
O E2     10.2.2.0/24 [110/20] via 10.1.1.2, 00:23:45, Ethernet1/0
      22.0.0.0/32 is subnetted, 1 subnets
B        22.22.22.22 [20/0] via 207.7.7.2, 00:15:28
      24.0.0.0/24 is subnetted, 1 subnets
B        24.2.4.0 [20/0] via 208.8.8.4, 00:16:23
      25.0.0.0/24 is subnetted, 1 subnets
B        25.2.5.0 [20/0] via 207.7.7.2, 00:15:28
      32.0.0.0/24 is subnetted, 1 subnets
B        32.3.2.0 [20/0] via 207.7.7.2, 00:15:28
      33.0.0.0/32 is subnetted, 1 subnets
B        33.33.33.33 [20/0] via 209.9.9.3, 00:14:58
      35.0.0.0/24 is subnetted, 1 subnets
B        35.3.5.0 [20/0] via 209.9.9.3, 00:14:58
      44.0.0.0/32 is subnetted, 1 subnets
B        44.44.44.44 [20/0] via 208.8.8.4, 00:16:23
      45.0.0.0/24 is subnetted, 1 subnets
B        45.4.5.0 [20/0] via 208.8.8.4, 00:16:23
      55.0.0.0/32 is subnetted, 1 subnets
B        55.55.55.55 [20/0] via 208.8.8.4, 00:14:20
      192.168.1.0/27 is subnetted, 1 subnets
O E2     192.168.1.128 [110/20] via 10.1.1.2, 00:23:45, Ethernet1/0
      207.7.7.0/24 is variably subnetted, 2 subnets, 2 masks
C        207.7.7.0/24 is directly connected, Ethernet1/2
L        207.7.7.1/32 is directly connected, Ethernet1/2
      208.8.8.0/24 is variably subnetted, 2 subnets, 2 masks
C        208.8.8.0/24 is directly connected, Ethernet1/1
L        208.8.8.1/32 is directly connected, Ethernet1/1
      209.9.9.0/24 is variably subnetted, 2 subnets, 2 masks
C        209.9.9.0/24 is directly connected, Ethernet1/3
L        209.9.9.1/32 is directly connected, Ethernet1/3
R1#
```

Ready   Telnet: 192.168.192.12‌9   69, 4   69 Rows, 113 Cols   Xterm   CAP NUM

R2

File Edit View Options Transfer Script Tools Window Help

P1 P2 A1 A2 S1 D1 D2 S2 R4 R3 R2 R1 I3 I2 I1 Google-DNS

```
R2>en
Password:
R2#
R2#config t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#Int loopback 0
R2(config-if)# ip add 2.2.2.2 255.255.255.255
% 2.2.2.2 overlaps with Loopback2
R2(config-if)#router ospf 1
R2(config-router)#router-id 2.2.2.2
R2(config-router)#network 2.2.2.2 0.0.0.0 area 0
R2(config-router)#network 10.1.1.4 0.0.0.3 area 0
R2(config-router)#network 10.1.1.0 0.0.0.3 area 0
R2(config-router)#interface e1/2
R2(config-if)#   ip ospf priority 255
R2(config-if)#end
R2#
R2#sh ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.1           0   FULL/DROTHER    00:00:32    10.1.1.1        Ethernet1/2
3.3.3.3           1   FULL/DR         00:00:35    10.1.1.6        Ethernet1/1
R2#
R2#sh ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.1           0   FULL/DROTHER    00:00:36    10.1.1.1        Ethernet1/2
3.3.3.3           1   FULL/DR         00:00:38    10.1.1.6        Ethernet1/1
R2#ping 8.8.8.8
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 8.8.8.8, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is 10.1.1.1 to network 0.0.0.0

O*E2  0.0.0.0/0 [110/1] via 10.1.1.1, 00:09:34, Ethernet1/2
      1.0.0.0/32 is subnetted, 1 subnets
O        1.1.1.1 [110/11] via 10.1.1.1, 00:32:34, Ethernet1/2
      2.0.0.0/32 is subnetted, 1 subnets
C        2.2.2.2 is directly connected, Loopback2
      3.0.0.0/32 is subnetted, 1 subnets
O        3.3.3.3 [110/11] via 10.1.1.6, 00:33:24, Ethernet1/1
      4.0.0.0/32 is subnetted, 1 subnets
O        4.4.4.4 [110/21] via 10.1.1.6, 00:33:24, Ethernet1/1
      10.0.0.0/8 is variably subnetted, 9 subnets, 3 masks
C        10.1.1.0/30 is directly connected, Ethernet1/2
L        10.1.1.2/32 is directly connected, Ethernet1/2
C        10.1.1.4/30 is directly connected, Ethernet1/1
L        10.1.1.5/32 is directly connected, Ethernet1/1
O        10.1.1.8/30 [110/20] via 10.1.1.6, 00:33:24, Ethernet1/1
O E2     10.1.4.4/30 [110/20] via 10.1.1.6, 00:23:41, Ethernet1/1
O E2     10.1.4.8/30 [110/20] via 10.1.1.6, 00:23:41, Ethernet1/1
O E2     10.2.1.0/24 [110/20] via 10.1.1.6, 00:23:41, Ethernet1/1
O E2     10.2.2.0/24 [110/20] via 10.1.1.6, 00:23:41, Ethernet1/1
      192.168.1.0/27 is subnetted, 1 subnets
O E2     192.168.1.128 [110/20] via 10.1.1.6, 00:23:41, Ethernet1/1
O     208.8.8.0/24 [110/30] via 10.1.1.6, 00:33:24, Ethernet1/1
R2#
```

Ready   Telnet: 192.168.192.12! 69, 4   69 Rows, 113 Cols   Xterm   CAP NUM

R3

File Edit View Options Transfer Script Tools Window Help

Enter host <Alt+R>

P1 P2 A1 A2 S1 D1 D2 S2 R4 R3 R2 R1 I3 I2 I1 Google-DNS

Session Manager

```
  Transmit Delay is 1 sec, State POINT_TO_POINT
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    oob-resync timeout 40
    Hello due in 00:00:02
  Supports Link-local Signaling (LLS)
  Cisco NSF helper support enabled
  IETF NSF helper support enabled
  Index 1/3/3, flood queue length 0
  Next 0x0(0)/0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 2
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1, Adjacent neighbor count is 1
    Adjacent with neighbor 4.4.4.4
  Suppress hello for 0 neighbor(s)
R3(config-if)#end
R3#sh ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
4.4.4.4           0   FULL/  -        00:00:32    10.1.1.10       Ethernet1/2
2.2.2.2           1   FULL/BDR        00:00:36    10.1.1.5        Ethernet1/1
R3#show ip EIGRP neighbors
R3#
R3#sh ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
4.4.4.4           0   FULL/  -        00:00:39    10.1.1.10       Ethernet1/2
2.2.2.2           1   FULL/BDR        00:00:35    10.1.1.5        Ethernet1/1
R3#ping 8.8.8.8
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 8.8.8.8, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is 10.1.1.5 to network 0.0.0.0

O*E2  0.0.0.0/0 [110/1] via 10.1.1.5, 00:08:55, Ethernet1/1
      1.0.0.0/32 is subnetted, 1 subnets
O        1.1.1.1 [110/21] via 10.1.1.5, 00:31:57, Ethernet1/1
      2.0.0.0/32 is subnetted, 1 subnets
O        2.2.2.2 [110/11] via 10.1.1.5, 00:32:45, Ethernet1/1
      3.0.0.0/32 is subnetted, 1 subnets
C        3.3.3.3 is directly connected, Loopback3
      4.0.0.0/32 is subnetted, 1 subnets
O        4.4.4.4 [110/11] via 10.1.1.10, 00:34:19, Ethernet1/2
      10.0.0.0/8 is variably subnetted, 9 subnets, 3 masks
O        10.1.1.0/30 [110/20] via 10.1.1.5, 00:32:45, Ethernet1/1
C        10.1.1.4/30 is directly connected, Ethernet1/1
L        10.1.1.6/32 is directly connected, Ethernet1/1
C        10.1.1.8/30 is directly connected, Ethernet1/2
L        10.1.1.9/32 is directly connected, Ethernet1/2
O E2     10.1.4.4/30 [110/20] via 10.1.1.10, 00:23:02, Ethernet1/2
O E2     10.1.4.8/30 [110/20] via 10.1.1.10, 00:23:02, Ethernet1/2
O E2     10.2.1.0/24 [110/20] via 10.1.1.10, 00:23:02, Ethernet1/2
O E2     10.2.2.0/24 [110/20] via 10.1.1.10, 00:23:02, Ethernet1/2
      192.168.1.0/27 is subnetted, 1 subnets
O E2     192.168.1.128 [110/20] via 10.1.1.10, 00:23:02, Ethernet1/2
O     208.8.8.0/24 [110/20] via 10.1.1.10, 00:34:19, Ethernet1/2
R3#
R3#
```

Ready   Telnet: 192.168.192.12!   69, 4   69 Rows, 113 Cols   Xterm   CAP NUM

```
P 10.1.1.0/30, 1 successors, FD is 281600
        via Redistributed (281600/0)
P 1.1.1.1/32, 1 successors, FD is 281600
        via Redistributed (281600/0)

R4#show ip EIGRP neighbors
EIGRP-IPv4 Neighbors for AS(100)
H   Address                 Interface              Hold Uptime   SRTT   RTO  Q  Seq
                                                   (sec)         (ms)       Cnt Num
1   10.1.4.6                Et1/0                    14 00:30:56 1025  5000  0  32
0   10.1.4.10               Et1/1                    14 00:30:56    7   100  0  21
R4#sh ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
3.3.3.3           0   FULL/  -        00:00:34    10.1.1.9        Ethernet1/2
R4#show ip EIGRP neighbors
EIGRP-IPv4 Neighbors for AS(100)
H   Address                 Interface              Hold Uptime   SRTT   RTO  Q  Seq
                                                   (sec)         (ms)       Cnt Num
1   10.1.4.6                Et1/0                    11 00:31:23 1025  5000  0  32
0   10.1.4.10               Et1/1                    14 00:31:23    7   100  0  21
R4#
R4#ping 8.8.8.8
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 8.8.8.8, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
R4#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is 10.1.1.9 to network 0.0.0.0

O*E2  0.0.0.0/0 [110/1] via 10.1.1.9, 00:11:55, Ethernet1/2
      1.0.0.0/32 is subnetted, 1 subnets
O        1.1.1.1 [110/31] via 10.1.1.9, 00:34:57, Ethernet1/2
      2.0.0.0/32 is subnetted, 1 subnets
O        2.2.2.2 [110/21] via 10.1.1.9, 00:35:45, Ethernet1/2
      3.0.0.0/32 is subnetted, 1 subnets
O        3.3.3.3 [110/11] via 10.1.1.9, 00:37:19, Ethernet1/2
      4.0.0.0/32 is subnetted, 1 subnets
C        4.4.4.4 is directly connected, Loopback4
      10.0.0.0/8 is variably subnetted, 10 subnets, 3 masks
O        10.1.1.0/30 [110/30] via 10.1.1.9, 00:35:45, Ethernet1/2
O        10.1.1.4/30 [110/20] via 10.1.1.9, 00:37:19, Ethernet1/2
C        10.1.1.8/30 is directly connected, Ethernet1/2
L        10.1.1.10/32 is directly connected, Ethernet1/2
C        10.1.4.4/30 is directly connected, Ethernet1/0
L        10.1.4.5/32 is directly connected, Ethernet1/0
C        10.1.4.8/30 is directly connected, Ethernet1/1
L        10.1.4.9/32 is directly connected, Ethernet1/1
D        10.2.1.0/24 [90/281856] via 10.1.4.10, 00:40:57, Ethernet1/1
                     [90/281856] via 10.1.4.6, 00:40:57, Ethernet1/0
D        10.2.2.0/24 [90/281856] via 10.1.4.10, 00:40:57, Ethernet1/1
                     [90/281856] via 10.1.4.6, 00:40:57, Ethernet1/0
      192.168.1.0/27 is subnetted, 1 subnets
D        192.168.1.128 [90/281856] via 10.1.4.10, 00:40:57, Ethernet1/1
                       [90/281856] via 10.1.4.6, 00:40:57, Ethernet1/0
      208.8.8.0/24 is variably subnetted, 2 subnets, 2 masks
C        208.8.8.0/24 is directly connected, Ethernet3/3
L        208.8.8.100/32 is directly connected, Ethernet3/3
R4#
```